# FEDERICO DOMINGUEZ MOLINA

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## EDUCATION

**The University of Chicago Computer Science Dept. & Harris School of Public Policy** — Chicago, IL
M.S. in Computational Analysis and Public Policy (MSCAPP) — June 2024

**Instituto Tecnológico Autónomo de México** — Mexico City
B.A. in Economics, Honors — June 2020

## PROFESSIONAL EXPERIENCE

**Energy & Environment Lab** — Chicago, IL
*Data Scientist* — June 2024 – Present

- Architected an automated global alert system to raise air pollution awareness using large language models and graph analysis, reaching over 250,000 impressions in three months
- Designed a social-media scraper extracting ~11 million daily posts, delivering $50,000 in monthly cost savings
- Led LLM classification improvement from 65% → 87% accuracy via prompt engineering — built annotated datasets and iterative evaluation loops for systematic prompt adjustment

**University of Chicago** — Chicago, IL
*Data Science Research Assistant* — February 2023 – June 2024

- Co-developed FastAPI web platform to play laboratory games and inform research on children's decision-making
- Created interactive visualizations and dashboards with Plotly to detect bias in 250+ policing research papers

**Coding it Forward** — Chicago, IL
*Data Engineering Fellow* — June 2023 – August 2023

- Automated data collection & cleaning for six websites with Python and reduced 10+ weekly hours of manual work
- Built a Python data platform allowing City of Long Beach's Climate Office to download data from 2010-2023, improving access and saving hundreds of work hours annually

**deep_dive** — Mexico City
*Senior Data Scientist* — October 2021 – October 2023

- Developed machine learning recommender system, using k-means and feature engineering, to offer customized loans to over 500,000 borrowers
- Created serverless computing architecture to collect data of over 250,000 commercial goods for strategic pricing

*Junior Data Scientist* — June 2020 – October 2021

- Extracted and parsed data from key Mexican governmental sources to support decision-making for 10+ clients
- Devised forecasting models using XGBoost, Random Forests and Linear Regression, along with an ensemble of the three, to become a top inflation forecaster in Mexico

## PERSONAL PROJECTS

**Trading Platform Maintenance** — November 2022 – September 2024

- Designed robust EC2 architecture and Python code to auto-trade over 10+ portfolios through Interactive Brokers, reducing trading errors by 95%
- Optimized AWS Cloud architecture by removing unused instances and stale data, reducing monthly costs by 35%
- Parallelized existing Python pipelines to extract price data for over 1,500 stocks, reducing runtime by 97% from two hours to under three minutes

## SKILLS

**Technical**: **Python** (transformers, pytorch, scikitlearn, xgboost, boto3, pyspark, sqlalchemy, streamlit, seaborn, matplotlib, plotly, fastapi, flask, pandas, numpy), **R**, **SQL, C, SAS, MATLAB, Power BI**
**Cloud Computing**: **AWS** (Lambda, EC2, Neptune, Route53, SQS, SNS, Glacier), **Google Cloud, Databricks**
**Publications**: *Academic Copaganda (w/ Dr. Robert Vargas)*